Compressed Air System - Maintenance and Leak Management

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Document type	Maintenance Manual	**Status**	Current
Effective date	2025-04-14	**Owner**	Facility Management
Classification	Internal		

1 Scope

This manual covers the two screw compressors K1 and K2 (each 75 kW), the refrigeration dryer and the compressed air network of the plant.

2 Operating values

Value	Target
Network pressure	6.5 bar (alarm below 6.0 bar)
Pressure dew point after dryer	+3 degrees C
Oil carry-over	max 0.01 mg/m3

3 Maintenance plan

Task	Interval
Check condensate drains	daily
Replace air intake filter	every 2000 operating hours
Replace oil separator cartridge	every 4000 operating hours
Oil change compressor	every 8000 operating hours
Leak detection round with ultrasonic detector	every 3 months

4 Leak management

Leaks are the biggest energy loss in the compressed air system. During the leak detection round, every leak is marked with a yellow tag and entered in the maintenance system with location and estimated size. Leaks are classified:

- Class A (audible, more than 10 l/min): repair within 1 week
- Class B (3 to 10 l/min): repair within 4 weeks
- Class C (below 3 l/min): repair at the next planned stop

A leak with a diameter of 3 mm at 6.5 bar loses roughly 10 l/s of air, which is about 3 kW of compressor power.

5 Revision history

Rev	Date	Change
A	2025-04-14	First issue